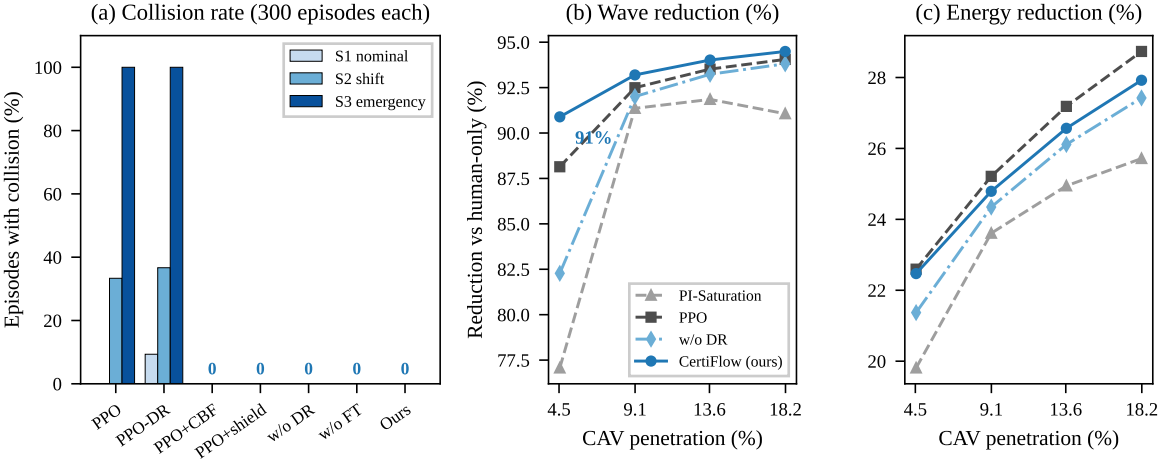
(a) Collision rate (300 episodes each)
Episodes with collision (%)
S1 nominal
S2 shift
S3 emergency
0
0
0
0
0
PPO
PPO-DR
PPO+CBF
PPO+shield
w/o DR
w/o FT
Ours
(b) Wave reduction (%)
Reduction vs human-only (%)
91%
PI-Saturation
PPO
w/o DR
CertiFlow (ours)
CAV penetration (%)
(c) Energy reduction (%)
CAV penetration (%)